



MVP Backend API Structure



AUTH FLOW (JWT + Session)

Flow:

1 User signup/login

2 Server returns:

- access_token (short life)
- refresh_token (long life)

3 Access token used for all requests

4 Refresh token used to renew session

All authenticated routes:

Authorization: Bearer <access_token>



AUTH ENDPOINTS

● POST /api/v1/auth/signup

Create account (pilot/admin by admin later)

Request

```
{
  "name": "Ali",
  "email": "ali@test.com",
  "password": "123456"
}
```

```
}
```

Response

```
{  
  "user_id": 12,  
  "access_token": "...",  
  "refresh_token": "..."  
}
```

POST /api/v1/auth/login

Request

```
{  
  "email": "user@mail.com",  
  "password": "123456"  
}
```

Response

```
{  
  "access_token": "...",  
  "refresh_token": "...",  
  "user": {  
    "id": 1,  
    "name": "Ali",  
    "role": "pilot"  
  }  
}
```

POST /api/v1/auth/refresh

Get new access token

```
{  
  "refresh_token": "..."  
}
```

POST /api/v1/auth/logout

Invalidate session

USER ENDPOINTS

GET /api/v1/users/me

Return current user profile

Response

```
{
  "id": 1,
  "name": "Ali",
  "role": "pilot"
}
```

MANUAL LIBRARY ENDPOINTS

GET /api/v1/manuals

List manuals (pilot/admin)

Response

```
[
  {
    "id": 1,
    "title": "A320 Manual",
    "size": 52300000,
    "created_at": "2026-01-01"
  }
]
```

GET /api/v1/manuals/:id/download

 Auth required

 Returns watermarked PDF (SERVER-SIDE generated)

No public URL.

POST /api/v1/manuals/upload (ADMIN)

Upload manual

Form-data:

file: pdf

title: string

Response:

```
{ "success": true }
```

DELETE /api/v1/manuals/:id (ADMIN)



FORM & SUBMISSION ENDPOINTS

GET /api/v1/forms/active

Returns current active form schema

```
{
  "form_id": 1,
  "version": 1,
  "fields": [
    {"name": "flight_no", "type": "text"},
    {"name": "description", "type": "textarea"}
  ]
}
```

POST /api/v1/submissions

Pilot submits form

Request

```
{
  "form_version_id": 1,
  "data": {
    "flight_no": "IR123",
    "description": "engine vibration"
  }
}
```

Optional file: multipart

Response

```
{
  "submission_id": 55,
  "hash_id": "ABC123XYZ",
  "status": "submitted"
}
```

GET /api/v1/submissions (ADMIN/VIEWER)

List all submissions

Pagination supported.

Response

```
{
  "page": 1,
  "total": 25,
  "items": [...]
}
```

GET /api/v1/submissions/:id

View single submission



ROLE ACCESS SUMMARY

Pilot:

- view manuals
- submit form

Admin:

- upload manuals
- view all submissions

Viewer:

- view submissions only

All enforced server-side.



ERROR HANDLING STANDARD

All errors:

```
{  
  "error": "message",  
  "code": 400  
}
```

Common codes:

- 400 bad request

- 401 unauthorized
 - 403 forbidden
 - 404 not found
 - 500 server error
-



PAGINATION STANDARD

Used in list endpoints:

Query:

?page=1&limit=20

Response:

```
{
  "page":1,
  "limit":20,
  "total":140,
  "items":[]
}
```



VERSIONING

All routes under:

/api/v1/

Future:

/api/v2/

So MVP stays stable.

OPENAPI / SWAGGER

I will generate Swagger after endpoints finalized.

Swagger includes:

- all endpoints
- request/response schema
- auth flow
- testing panel

Frontend & backend will both use it.